Ex. No: 6	Setting and configuring two factor authentication in the Mobile phone.
Date:	

Aim:

To enable and configure Two-Factor Authentication (2FA) on a mobile phone to enhance account security.

Objective

- To understand the concept of Multi-Factor Authentication (MFA).
- To implement 2FA using SMS and Authenticator App methods.
- To analyze the security benefits of 2FA.

System Requirements

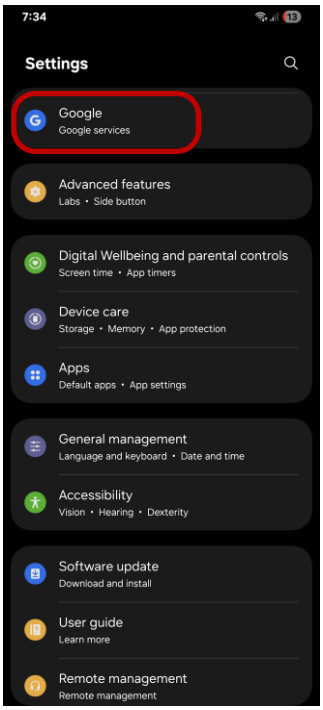
- Android / iOS Smartphone
- Active Internet Connection
- Google Account / Email Account
- Authenticator App (optional)

Procedure:

1.Open Settings on the mobile phone.

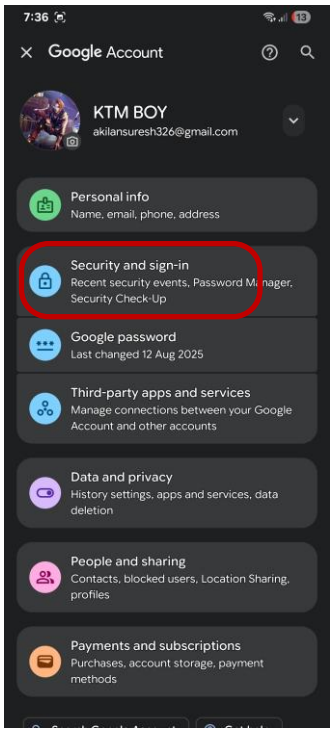
- Unlock the mobile device and open the **Settings** application from the home screen or app drawer.

2. Tap Google → Manage your Google Account.



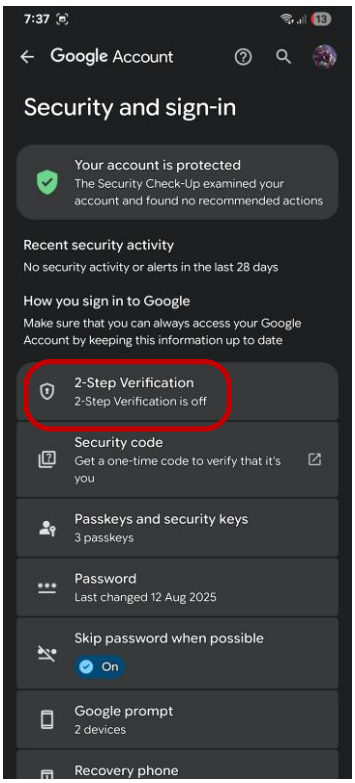
3 .Go to Security tab.

- Inside the Google Account page, navigate to the **Security** tab which contains options related to account protection and login settings.



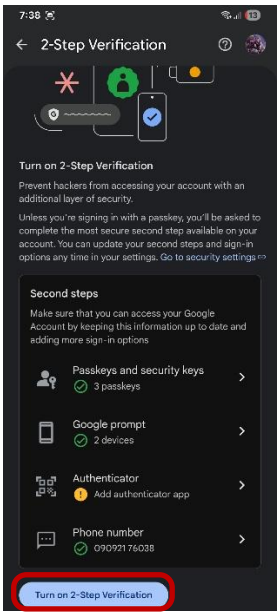
4.Under “Signing in to Google”, select 2-Step Verification.

- Scroll down to the **Signing in to Google** section and choose **2-Step Verification** to add an extra layer of security to the account.



5.Click Get Started.

- Tap the **Get Started** button to begin the setup process for enabling Two-Factor Authentication.

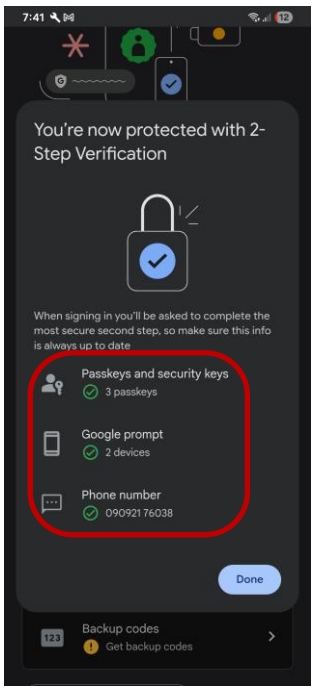


6.Enter account password.

- For security verification, enter the **Google account password** to confirm the identity of the user.

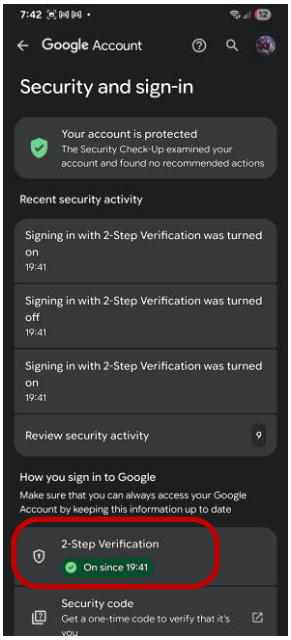
7. Choose verification method:

- SMS OTP
- Google Prompt
- Authenticator App



8. Confirm and Turn ON 2-Step Verification.

- After selecting the verification method, confirm the setup and enable **2-Step Verification** to secure the Google account.



Result

Two-Factor Authentication successfully enabled for Google account.

Part B: Configuring Authenticator App (More Secure Method)

Using Google Authenticator

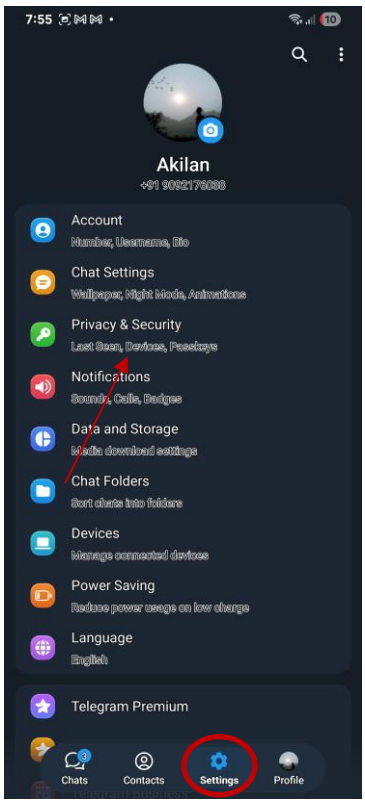
Procedure (Android / iOS)

Step 1: Open Telegram

- Launch Telegram app on mobile phone.

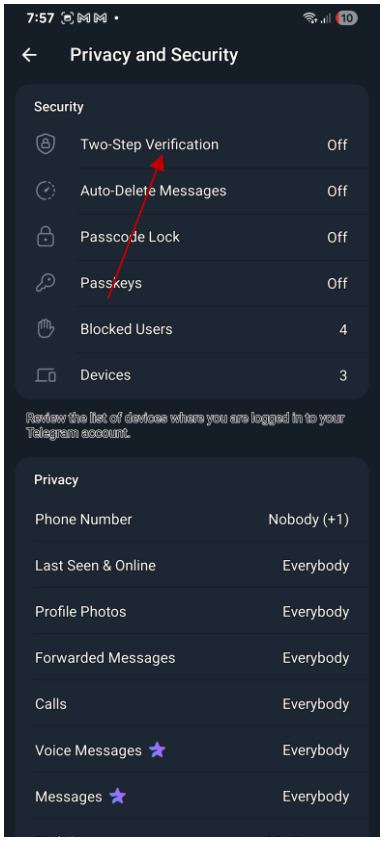
Step 2: Go to Settings

- Tap on Menu (☰)
- Select Settings



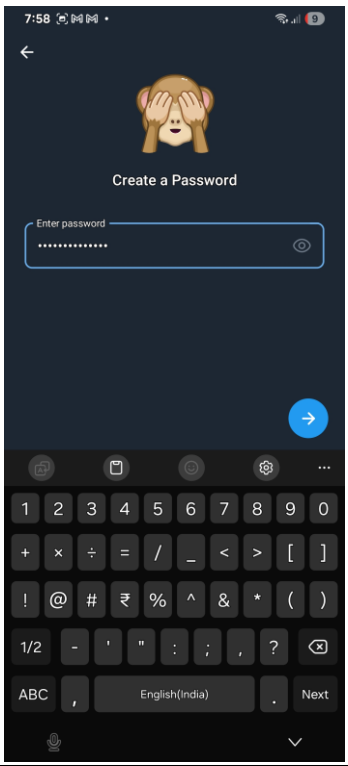
Step 3: Open Privacy and Security

- Tap Privacy and Security



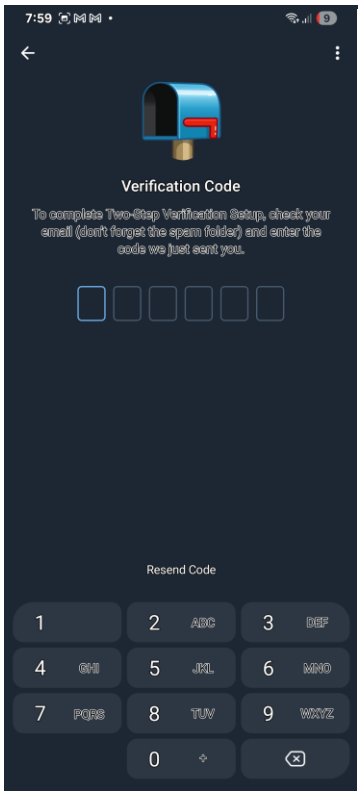
Step 5: Set Password

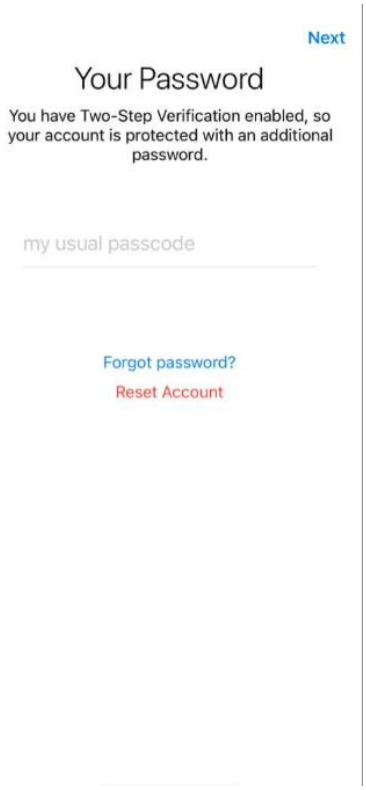
- Tap Set Password
- Enter a strong password
- Re-enter to confirm
- Tap Continue

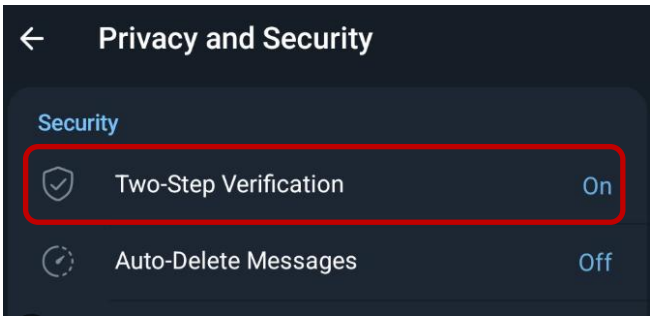


Step 6: Add Recovery Email

- Enter recovery email address
- Verify email using confirmation code
- Tap Done







Password Policy Recommendation

- Minimum 8–12 characters
- Include uppercase, lowercase, numbers, symbols
- Avoid personal information

Example: Tg@2026#Secure

Security Analysis

Scenario	Without 2FA	With 2FA
SIM Swap Attack	Account compromised	Protected by cloud password
OTP Interception	Account hacked	Cannot login without second password
Device Theft	Risky	Additional password required

Result

Two-Step Verification successfully enabled in Telegram.